

RM04

Coding-Decoding

1-3 questions in every RAS Prelims paper without exception - 2013, 2016, 2018, 2021, 2023 and 2024 all carry at least one. Pure method, no knowledge, no calculation beyond counting to 26. This chapter should never be left unattempted.

A code is a rule, and there are only seven rules RPSC ever uses. The whole chapter is: identify which of the seven you are looking at, verify it on a second letter or a second word, then apply it. Students lose these marks not because the codes are hard but because they apply a shift method to a substitution question.

Coding-Decoding

Letter coding - shifts

- Constant forward +1, +2, +3
- Constant backward -1, -2, -3
- Alternating +1, -1
- Shift by place in the word (+1, +2, +3)

Structural - no shift at all

- Full reversal DELHI to IHLED
- Pair interchange MONKEY to OMKNYE
- First-last swap SYSTEM to MYSTES
- Opposite letters DOG to WLT
- Reverse then shift CAT to UBD

Number coding

- Position written out CAT = 3-1-20
- Sum of positions CAT = 24
- Multiple of position A=2, B=4
- Reverse position Z=1 ... A=26
- Position plus or minus a constant

Mixed letter-digit

- Build a letter-to-digit dictionary
- Shared letters must agree
- Disagreement means mis-alignment

Substitution and sentences

- 'X is called Y' - answer in plain language first
- Two sentences sharing one word fix that pair
- Cross out each code word as it is fixed

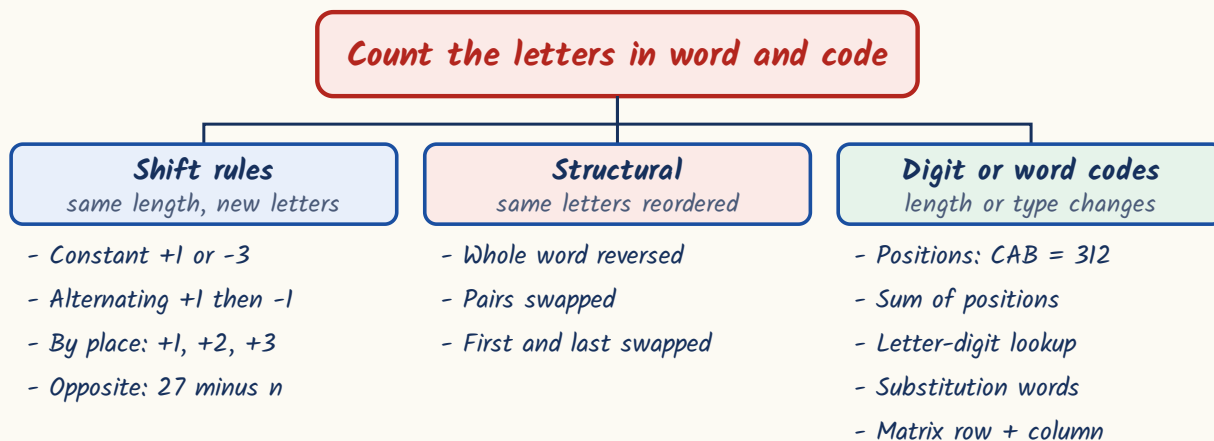
Conditional and matrix

- Plain code first, then conditions in order
- Watch first and last letter for vowels
- Matrix: row number then column number
- Note which letter is left out

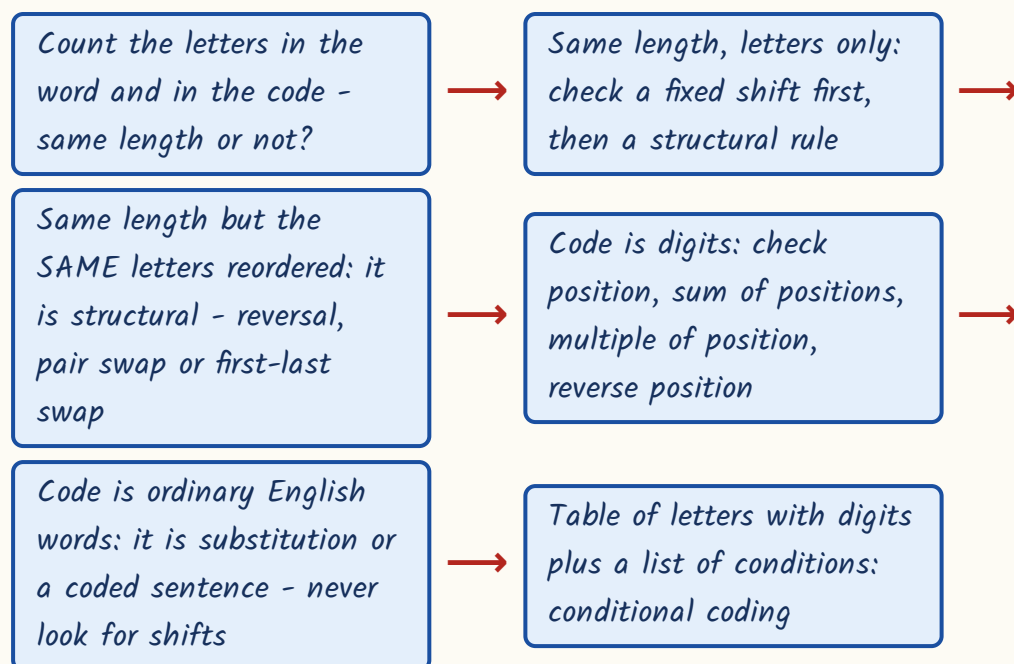
The first question to ask, every single time

Before you touch the letters, count. Is the code the same length as the word? If yes, you are looking at a letter-to-letter rule and your job is to find the shift or the rearrangement. If the code is shorter, or is a number, or is an ordinary English word, then it is number coding, substitution or compression - a completely different method. Getting this one decision right in five seconds saves you the minute students waste hunting for a shift that does not exist.

Diagnose the code before you solve it



Decide the type before you solve



The seven rule families

Family	Signature	Example
1. Constant forward shift	Every letter moves the same number forward	MADRAS to NBESBT (+1)
2. Constant backward shift	Every letter moves the same number back	TRIPPLE to SQHOOKD (-1)
3. Alternating shift	+1 then -1 then +1	CHAIR to DGBHS
4. Shift by place in word	First +1, second +2, third +3	ABC to BDF
5. Structural, no shift	Same letters, new order	DELHI to IHLED; MONKEY to OMKNYE
6. Opposite letter	27 minus the position	DOG to WLT
7. Combination	Reverse first, then shift	CAT to UBD

YAAD RAKHO

Remember: you cannot do this chapter without $A=1$ to $Z=26$ in your head, plus EJOTY (E5, J10, O15, T20, Y25) and CFILORUX (C3, F6, I9, L12, O15, R18, U21, X24). And the opposite-letter rule, 27 minus the position, giving A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N. Learn these before you learn any coding technique at all.

Constant shift coding - the core method

This is the family that appears most often. Write the word above and the code below, letter under letter. Compute the gap for the first letter. Then compute it again for the second and third. If the gap is the same, you have the rule; apply it to the asked word. If the gap changes sign or size, you are in family 3 or 4, not family 1.

Three steps, never skip the middle one

Find the shift on ONE pair of letters



Verify the same shift on a SECOND and THIRD pair



Apply it to every letter of the asked word, wrapping round the alphabet at Z and A

- MADRAS is written as NBESBT. How is BOMBAY written?
- M to N is +1. A to B is +1. D to E is +1. R to S, A to B, S to T - all +1. Rule fixed.
- Apply +1 to BOMBAY: B to C, O to P, M to N, B to C, A to B, Y to Z.
- Answer: CPNCBZ. Note the last letter - $Y + 1 = Z$, and the option CPNCAZ that changes the fifth letter is the planted trap.

ASKED IN RAS

Asked in RAS Pre 2016 - MADRAS is coded NBESBT, so BOMBAY is CPNCBZ. This exact +1 pattern has repeated across RAS and RPSC papers. If you see a code whose first letter is one after the word's first letter, test +1 on all six letters immediately.

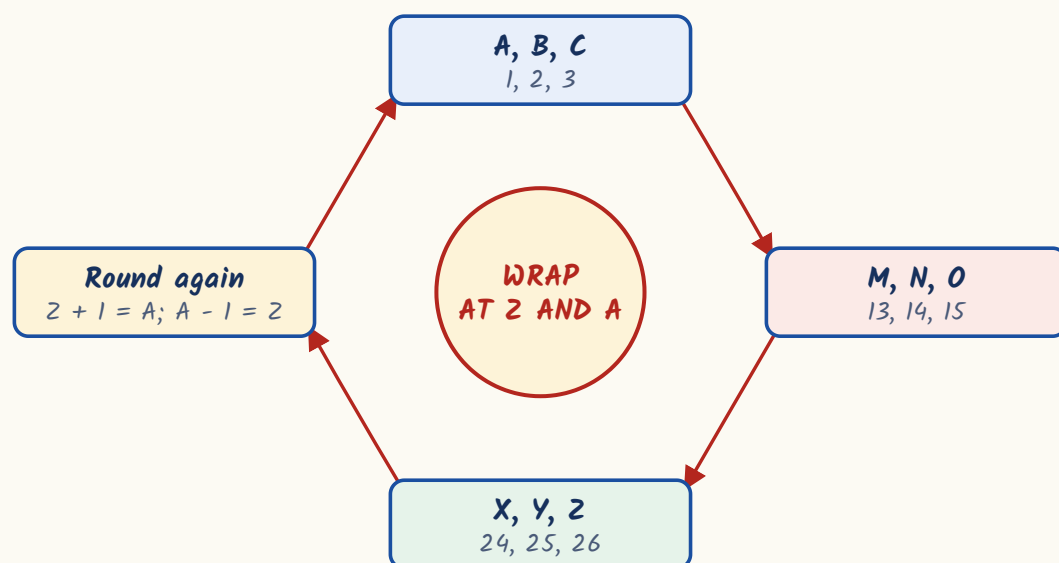
- FRIEND is written as HTKGPF. How is CANDLE written?
- F to H is +2, R to T is +2, I to K is +2, E to G, N to P, D to F - all +2.
- Apply +2 to CANDLE: C to E, A to C, N to P, D to F, L to N, E to G.
Answer: ECPFNG.
- TEACHER to WHDFKHU is the same family with +3, so STUDENT becomes VWXGHQW.
- TRIPPLE to SQHOOKD is -1, so DISPOSE becomes CHRONRD. Backward shifts are just as common as forward ones.

ASKED IN RAS

Asked in RAS Pre 2021 - FRIEND is coded HTKGPF, so CANDLE is ECPFNG. The nearest wrong option was ECPFNH, differing only in the last letter. Always compute the last letter of your answer separately and check it against the options first.

- WATER is coded TXQBO. How is FIRE coded?
- W(23) to T(20) is -3. A(1) to X(24) - here you must wrap: A minus 3 goes back past A to X. Still -3.
- T(20) to Q(17) is -3, E(5) to B(2) is -3, R(18) to O(15) is -3. Rule confirmed on five letters.
- Apply -3 to FIRE: F to C, I to F, R to O, E to B. Answer: CFOB.
- Wrapping is where marks are lost. Treat the alphabet as a circle: $A - 1 = Z$, $A - 2 = Y$, $A - 3 = X$; $Z + 1 = A$, $Z + 2 = B$.
- DECODING a message means running the rule backwards. HELP is coded GDKO, i.e. -1; so to read RTMCZX you apply +1.
- R to S, T to U, M to N, C to D, Z to A, X to Y - the message is SUNDAY. Reverse the sign of the shift, never the word.

The alphabet is a ring, not a line



Structural rules - when nothing shifts at all

Here is the fastest diagnostic in the chapter. Look at the letters of the code and the letters of the word. Are they the SAME letters, just in a different order? If yes, stop looking for a shift - it is a structural rule, and there are only three common ones plus one that combines with a shift. Check them in the order given in the table.

Structural rules and how to spot them

Rule	Test	Example
Full reversal	Code read backwards is the word	DELHI to IHLED
Successive pair interchange	Letters 1-2 swap, 3-4 swap, 5-6 swap	MONKEY to OMKNYE
First and last interchanged	Only the two end letters move	SYSTEM to MYSTES
Opposite letters (27 - n)	Letters are different, but each pair sums to 27	DOG to WLT
Reverse, then shift	Letters differ, but the reversed word is a fixed shift away	CAT to UBD

- DELHI is written as IHLED. How is MUMBAI written? Read the word backwards: I-A-B-M-U-M. Answer: IABMUM.
- MONKEY is written as OMKNYE - check: MO becomes OM, NK becomes KN, EY becomes YE. Successive pairs swapped.
- So CHAIRS becomes CH to HC, AI to IA, RS to SR. Answer: HCIASR.
- SYSTEM is coded MYSTES - only the M and the S at the ends have traded places. So NUMBER becomes RUMBEN.
- The pair-interchange trap: HCAIRS keeps the middle pair unswapped. Check EVERY pair, not just the first.

- DOG is written as WLT. Different letters, so test opposites: D(4) and W(23) sum to 27. O(15) and L(12) sum to 27. G(7) and T(20) sum to 27.
- So the rule is 'replace each letter by 27 minus its position'.
- CAT: C(3) becomes 24 = X, A(1) becomes 26 = Z, T(20) becomes 7 = G. Answer: XZG.
- Combination rule: CAT is written as UBD. Reverse CAT to get TAC, then shift +1: T to U, A to B, C to D.
- Apply to DOG: reverse to GOD, then +1 gives H, P, E. Answer: HPE. Test 'reverse then shift' whenever a plain shift fails but the letter-gaps look regular in reverse order.

ASKED IN RAS

Asked in RAS Pre 2023 - DOG is coded WLT, so CAT is XZG. This is the opposite-letter family and it is a repeat item. The moment you see a three-letter code where the letters sit in the far half of the alphabet from the word, add each pair and look for 27.

Alternating shifts and shift-by-position

These two families exist to punish the student who checked only one letter. Both begin with a +1 on the first letter, so both look like a simple forward shift until you reach letter two. Whenever your +1 rule fails on the second letter, do not abandon the question - test these two before anything else.

- CHAIR is written as DGBHS. How is TABLE written?
 - C to D is +1. H to G is -1. A to B is +1. I to H is -1. R to S is +1. The shift alternates +1, -1.
 - Apply to TABLE: $T+1 = U$, $A-1 = Z$ (wrapping), $B+1 = C$, $L-1 = K$, $E+1 = F$.
 - Answer: UZCKF. The wrap at A is exactly where the wrong option UZBKF is aimed.
-
- ABC is coded as BDF. How is PQR coded?
 - A to B is +1, B to D is +2, C to F is +3. The shift equals the letter's place in the word.
 - Apply to PQR: $P+1 = Q$, $Q+2 = S$, $R+3 = U$. Answer: QSU.
 - For a longer word this continues: fourth letter +4, fifth +5. Number the letters 1, 2, 3, 4 above the word before you start.
 - Distinguish from family 1: in a constant shift the gap is the same for every letter; here it grows by exactly one each time.

TRAP

Never fix a rule from the first letter alone. A +1 constant shift, an alternating +1/-1 shift and a shift-by-position all produce exactly the same first letter, and a full reversal agrees with a pair interchange on plenty of words. Compute letter two and letter three before you commit, and where two words are given, verify on the second word as well. This single habit is worth more marks in this chapter than any mnemonic.

Number coding – five forms, and how to tell them apart

When the code is digits, ask how many digits there are. A code with one group per letter is positional. A single small number for the whole word is a sum. A code where every value is even and double a familiar position is a multiple. A code where A is large and Z is small is reverse position. Test the form on the given word, then confirm it on a second given word if one exists - one word alone can fit several rules.

The five forms of number coding

Form	Rule	Example
Position written out	Each letter becomes its position	CAT = 3-1-20; MANGO = 13-1-14-7-15
Positions run together	Positions concatenated with no separator	CAT = 3120; DOG = 4157
Sum of positions	Add the positions of all letters	CAT = $3+1+20 = 24$
Multiple of position	$A = 2, B = 4, C = 6 \dots$ (twice the position)	FACE = $12-2-6-10$
Reverse position	$Z = 1, Y = 2 \dots A = 26$, i.e. 27 minus n	CAB = $24-26-25$

- CAB is coded 312 with $A=1, B=2 \dots Z=26$. What is FED? Positions: $F=6, E=5, D=4$. Answer: 654.
- MANGO is coded 13-1-14-7-15, so APPLE is $A=1, P=16, P=16, L=12, E=5$, giving 1-16-16-12-5.
- CAT = 3120 and DOG = 4157 - here the positions are simply written one after another with no gaps.
- So BAT is $B(2) A(1) T(20) = 2120$. Watch the option 2121, which is the planted near-miss.

ASKED IN RAS

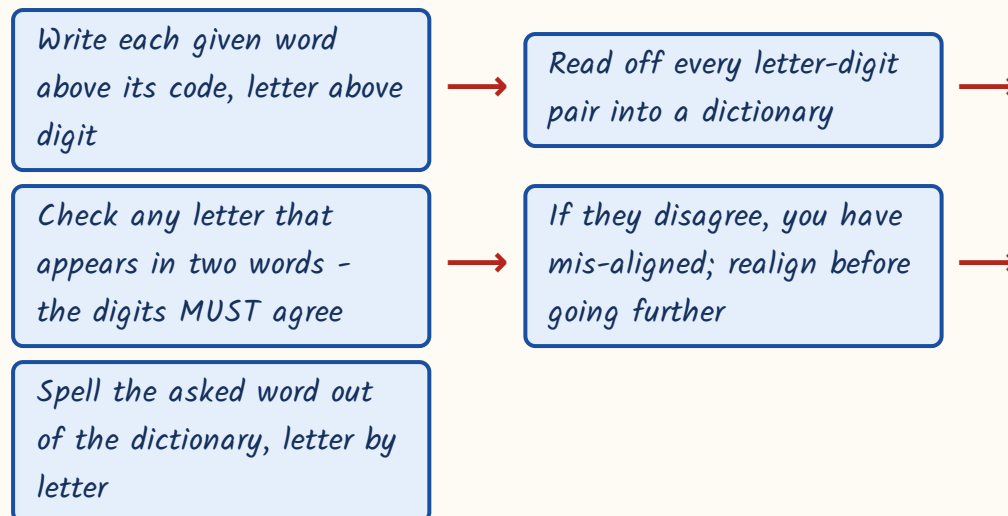
Asked in RAS Pre 2018 - with $A = 1$ to $Z = 26$, CAB is coded 312, so FED is coded 654. This is the simplest coding item RPSC sets, and it is free marks provided the alphabet table is genuinely memorised.

- SUN is written as 54. What is MOON? Sum test: $S(19) + U(21) + N(14) = 54$. Confirmed.
- MOON: $M(13) + O(15) + O(15) + N(14) = 57$. Answer: 57.
- CAT = 24, since $3 + 1 + 20 = 24$. So DOG = $4 + 15 + 7 = 26$.
- LOVE = 54, since $12 + 15 + 22 + 5 = 54$. So LIFE = $12 + 9 + 6 + 5 = 32$.
- If $A = 2$, $B = 4$, $C = 6$, each code is twice the position: FACE = 12-2-6-10.
- If the letters are numbered from the end, $Z = 1$ up to $A = 26$, then reverse position = $27 - n$, so CAB = 24-26-25.

Mixed letter-digit coding - build the dictionary

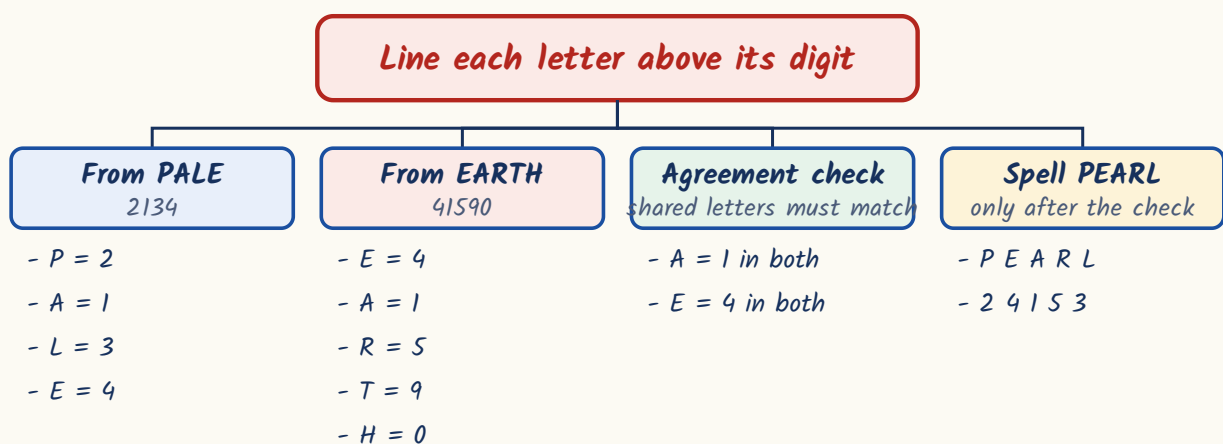
Here you are given two or three words with their number codes and asked to code a fourth. There is no shift and no arithmetic. Each letter simply owns a digit. Your job is to build the letter-to-digit dictionary by lining up the words with their codes, and then to spell out the asked word from that dictionary. The letters shared between the given words are your proof that you have lined them up correctly.

Dictionary method



- PALE = 2134 and EARTH = 41590. Code PEARL.
- From PALE: P=2, A=1, L=3, E=4. From EARTH: E=4, A=1, R=5, T=9, H=0.
- Consistency check: A appears in both and is 1 in both; E appears in both and is 4 in both. The alignment is right.
- Spell PEARL: P=2, E=4, A=1, R=5, L=3. Answer: 24153.

Building the dictionary: PALE = 2134, EARTH = 41590



- ROSE = 6821, CHAIR = 73456, PREACH = 961473. Code SEARCH.
- From ROSE: R=6, O=8, S=2, E=1. From CHAIR: C=7, H=3, A=4, I=5, R=6 - and R=6 agrees with ROSE, so the alignment holds.
- From PREACH: P=9, and R=6, E=1, A=4, C=7, H=3 all agree with what we already have.
- Spell SEARCH: S=2, E=1, A=4, R=6, C=7, H=3. Answer: 214673.
- Do the agreement check BEFORE you spell the answer - it costs ten seconds and it is the only thing standing between you and a wrong answer.

TRAP

If a shared letter gets two different digits, do not average, guess or ignore it. You have mis-aligned the word against its code - most often by starting from the wrong end, or by mis-counting a repeated letter such as the double O in MOON. Rewrite the alignment from scratch.

Substitution coding and coded sentences

Shift-based coding vs substitution coding - do not mix the methods

Shift / letter coding	Substitution coding
The code is a meaningless letter string (NBESBT)	The code words are ordinary English words (sea, water, blue)
You look for an arithmetic rule on letter positions	There is no arithmetic at all - it is a renaming list
Method: find the shift, verify it, apply it	Method: answer in plain language first, then translate the answer
Verification is on a second letter pair	Verification is on a second sentence sharing one word
Wrong answer comes from the alphabet wrap	Wrong answer comes from giving the REAL word instead of its new name

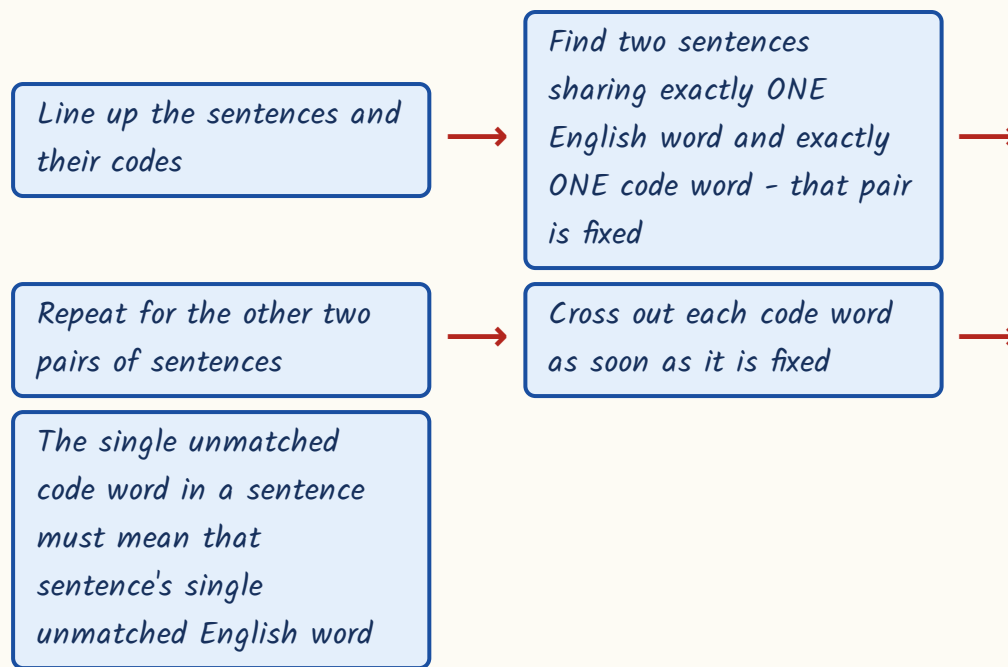
Substitution questions read 'if X is called Y, and Y is called Z, then ...'. The trick is to answer the question honestly in ordinary English first, and only then look up what that answer is called in the new language. Do it in that order and the question becomes trivial. Do it in the other order and you will pick the real-world word, which is always sitting there as an option.

- If 'sky' is called 'sea', 'sea' is called 'water', 'water' is called 'air', 'air' is called 'cloud' - where do fish live?
- Step 1, plain English: fish live in water.
- Step 2, translate: in this language 'water' is called 'air'. Answer: air.
- Same drill: if 'red' is called 'yellow', blood is really red, so the answer is yellow.
- If 'pen' is called 'book', we write with a pen, so the answer is book.
- If 'blue' means 'dance', a person dances to music, so the answer is blue.

ASKED IN RAS

Asked in RAS Pre 2013 - the sky/sea/water/air chain, asking where fish live. Answer: air. Note that 'water' appears as an option and is the answer to the plain-English step, not to the question. Never stop after step 1.

Coded sentences - the cross-out method



- 'tim nap sok' = 'she is intelligent'; 'nap dor lin' = 'is very smart'; 'sok lin bap' = 'intelligent smart boy'. Which word means 'boy'?
- Sentences 1 and 2 share only the word 'is' and only the code 'nap', so nap = is.
- Sentences 1 and 3 share only 'intelligent' and only 'sok', so sok = intelligent.
- Sentences 2 and 3 share only 'smart' and only 'lin', so lin = smart.
- In sentence 3 the only code left is 'bap' and the only English word left is 'boy'. Answer: bap.
- Same structure: 'ma po ri' / 'po lin sa' / 'ri sa dun' gives po = eat, ri = fish, sa = bones, so lin = dogs.

ASKED IN RAS

Asked in RAS Pre 2024 - 'gel hai jar' means 'apples are ripe' and 'hai tir sup' means 'children are playing'; which word means 'are'? Answer: hai. Only one English word is common and only one code word is common, so the pair is forced. Two sentences sharing exactly one word is the fastest item in the whole paper.

Conditional coding, matrix coding and coding by analogy

Conditional coding gives you a small letter-to-digit table and then two or three conditions that override it. The method is mechanical: write the plain digit code first, ignoring all conditions. Then apply each condition in the order printed. Then look again at the first and the last letter of the group, because almost every condition is about those two and whether they are vowels. Underline the first and last letter before you begin.

- Table: P=4, M=7, A=2, K=9, R=5, E=3, T=8.
- Condition (i): if the first letter is a vowel, both the first and the last letter become \$.
- Condition (ii): if the last letter is a vowel and the first is not, the last letter becomes #.
- Code AMKRT: plain code is 2-7-9-5-8. A is a vowel and it is first, so condition (i) fires - first and last both become \$. Answer: \$795\$.
- Code PMAKE: plain code is 4-7-2-9-3. P is not a vowel so (i) does not fire; E is a vowel and last, so (ii) fires. Answer: 4729#.
- Code KRAMP: plain code is 9-5-2-7-4. Neither K nor P is a vowel, so no condition applies. Answer: 95274.
- Notice the third case - 'no condition applies' is a real answer and RPSC uses it to catch students who assume a condition must fire.

- Table: B=6, J=1, N=4, S=9, U=3, D=7, I=2.
- Condition (i): if BOTH the first and the last letter are consonants, the first letter becomes *.
- Condition (ii): if the first is a vowel and the last a consonant, their two codes are interchanged.
- Code UDNSB: plain code is 3-7-4-9-6. U is a vowel, B is a consonant, so (ii) fires - swap the first and last codes. Answer: 67493.
- Code SNBDJ: plain code is 9-4-6-7-1. S and J are both consonants, so (i) fires and the first code becomes *. Answer: *4671.
- Vowels for these questions are A, E, I, O, U only. Y is treated as a consonant unless the question says otherwise.
- If two conditions could both apply, the paper normally states the precedence; if not, apply condition (i) first.

Matrix coding gives you a grid of letters and codes each letter by its row number followed by its column number. Two things to check before you start: whether the rows or the columns are numbered first, and which letter has been left out of the grid - it is usually Z, and knowing it is missing stops you miscounting the later letters.

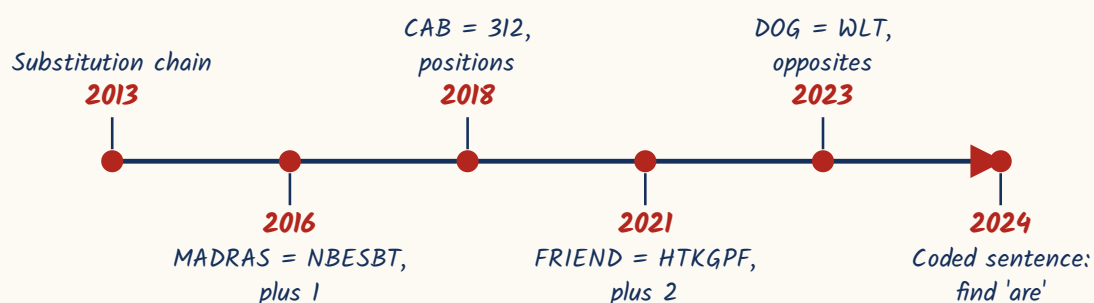
- Grid with rows R0 to R4 and columns C0 to C4, filled A to Y (Z not used):
row 0 is A B C D E, row 1 is F G H I J, row 2 is K L M N O, row 3 is P Q R S T, row 4 is U V W X Y.
- Code CAT: C is row 0, column 2 = 02; A is row 0, column 0 = 00; T is row 3, column 4 = 34. Answer: 02-00-34.
- Code MOON: M = 22, O = 24, O = 24, N = 23. Answer: 22-24-24-23.
- Decode 10-13-32-04: row 1 column 0 = F, row 1 column 3 = I, row 3 column 2 = R, row 0 column 4 = E. The word is FIRE.
- The commonest error is reading 20 as row 0 column 2 - always read ROW first unless the question says otherwise.

- Coding by analogy - BOOK : CPPL :: LOCK : ? Verify on every letter: B to C, O to P, O to P, K to L, all +1.
- Apply +1 to LOCK: M, P, D, L. Answer: MPDL.
- GATE : HBUF :: DOOR : ? The rule is +1 on each letter, so DOOR becomes EPPS.
- MOTHER : PRWKHU :: FATHER : ? The rule is +3, so FATHER becomes IDWKHU.
- LIGHT : JGEFR :: SOUND : ? The rule is -2, so SOUND becomes QMSLB.
- CAT : XZG :: DOG : ? The rule is the opposite letter, so DOG becomes WLT.
- The method is identical to ordinary letter coding - the analogy format changes nothing except the layout.

YAAD RAKHO

Remember the fastest elimination in this chapter: once you have the rule, do not code the whole word. Code only the FIRST and the LAST letter of the answer and compare against the options. Three options usually die at once, and where two survive, one more letter settles it. This turns a ninety-second item into a twenty-second one.

Coding-decoding in past RAS papers



REVISION RECAP

1. First question always: is the code the same length as the word? That single check picks the family.
2. Same letters in a new order means structural - reversal, pair interchange or first-last swap. No shift to find.
3. Constant shift method: find the shift on one pair, VERIFY on a second and third, then apply.
4. MADRAS to NBESBT is +1, so BOMBAY is CPNCBZ. FRIEND to HTKGPF is +2, so CANDLE is ECPFNG.
5. Backward shifts are as common as forward ones: TRIPPLE to SQHOOKD is -1; WATER to TXQBO is -3.
6. Wrap the alphabet: $A - 3 = X$, $Z + 2 = B$. This is where most marks are dropped.
7. Alternating shift: CHAIR to DGBHS is +1, -1, +1, -1, so TABLE is UZCKF.
8. Shift by place in the word: ABC to BDF, so PQR is QSU.
9. Opposite letter = 27 minus position: DOG to WLT, CAT to XZG.
10. Reverse then shift: CAT to UBD means reverse to TAC then +1; so DOG gives HPE.
11. DELHI to IHLED is full reversal; MONKEY to OMKNYE is pair interchange; SYSTEM to MYSTES is first-last swap.
12. Number coding forms: positions written out, positions concatenated, sum of positions, twice the position, reverse position.
13. Sum coding must be confirmed on a second word - one word alone fits several rules. $SUN = 54$ gives $MOON = 57$.
14. Mixed coding: build a letter-to-digit dictionary; shared letters must agree, and disagreement means mis-alignment.
15. $PALE = 2134$ with $EARTH = 41590$ gives $PEARL = 24153$.
16. Substitution: answer in plain English first, then translate. Fish live in water; water is called air, so answer air.
17. Coded sentences: two sentences sharing exactly one English word and one code word fix that pair. Cross out as you go.
18. Conditional coding: plain code first, conditions in the printed order, then re-check first and last letters - and 'no condition applies' is a genuine answer, KRAMP stays 95274.
19. Matrix coding: row number first, then column; note that Z is usually left out of the grid.
20. Elimination shortcut: code only the first and last letter of the answer and three options die at once.